

Statistical Design and Models for Evaluation Studies (37495)

Lecture 2 - Completely Randomised Designs

Lecturer: Joanna Wang

School of Mathematical and Physical Sciences, UTS

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1. Lecture outline

Topics:

- Example 1
- Mathematical setup
- ANOVA
- Analysis in R
- Example 2
- Example 3

Completely randomised designs

- We will develop the design and tests necessary to compare a treatments or treatment levels
- Assume there is no structural relationship among the treatments
- Assume we have a set of identical experimental units on which to compare the treatments

Example 1

An experiment was conducted to compare how different tasks affect a worker's pulse rate. In this experiment there were 60 male workers who were randomly assigned to the 6 tasks such that each task had 10 workers assigned to it. Each group was trained to perform their assigned task. On one day after training each group performed their assigned task for one hour and then the pulse rates of the workers in each group were measured. Is there a significant difference in the pulse rates for the 6 tasks?

Example 1

The results from this experiment are:

Task 1		Task 2		Task 3		Task 4		Task 5		Task 6	
27	31	29	28	34	36	34	34	28	28	28	26
26	32	37	24	34	41	42	44	26	35	29	25
39	37	35	40	30	44	40	47	31	30	35	34
38	39	40	31	44	32	34	31	34	34	37	28
30	28	30	25	32	31	45	28	26	20	21	28

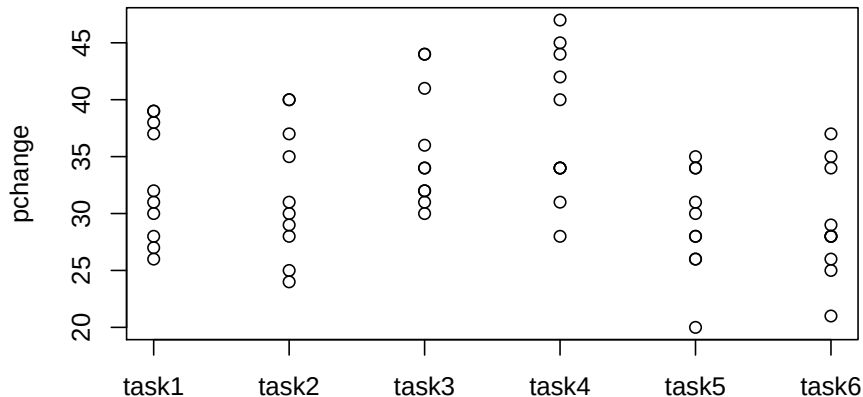
Example 1

Start with a scatter plot...

```
# Clear stored objects.
rm(list=ls()) # remove all objects
# Setup data
obs <- c(27,31,29,28,34,36,34,34,28,28,28,26,26,32,37,24,34,41,42,
        44,26,35,29,25,39,37,35,40,30,44,40,47,31,30,35,34,38,39,
        40,31,44,32,34,31,34,34,37,28,30,28,30,25,32,31,45,28,26,
        20,21,28) #response data
treatments <- rep(rep(c("task1","task2","task3","task4","task5",
"task6"),each=2),5) #treatment data
pulse <- data.frame(obs, treatments)
names(pulse) <- c("pchange", "task")
```

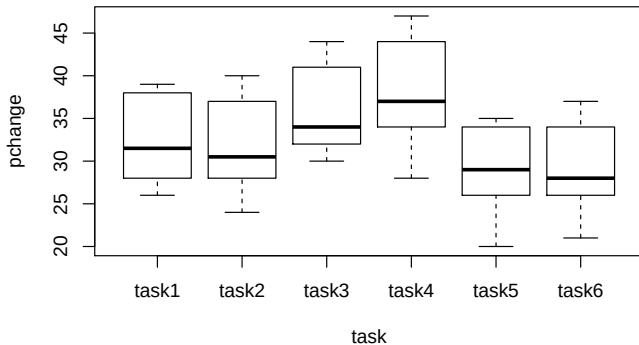
Example 1

```
stripchart(pchange~task, vertical=TRUE, pch=1, data=pulse)
```



Example 1

```
boxplot(pchange~task, data=pulse)
```



It looks like there may be some significant differences in the means.

Mathematical model

- The response for each of the a treatments is the random variable of interest, say y
- Denote y_{ij} the j th observation ($j = 1, \dots, n$) receiving i th treatment ($i = 1, \dots, a$)
- We have a independent samples (one sample from each of the treatments)
- *Means model*

$$y_{ij} = \mu_i + \epsilon_{ij}, \quad \text{for } i = 1, \dots, a \text{ and } j = 1, \dots, n,$$

where μ_i is the mean response of the i th treatment.

- *Effects model*

$$y_{ij} = \mu + \tau_i + \epsilon_{ij}, \quad \text{for } i = 1, \dots, a \text{ and } j = 1, \dots, n,$$

$$\begin{aligned}\mu_i &= \mu + \tau_i \\ \tau_i &= \mu_i - \mu\end{aligned}$$

where the mean response μ_i is written as an overall mean and the i th treatment effect, τ_i . Note that $\sum_{i=1}^a \tau_i = 0$. μ

- Here we assume there are n observations made on each of the treatments. Similar models can be defined for unequal sample sizes, ie $n = n_i$
- ϵ_{ij} is random error component. As usual for errors, we assume that the random variables ϵ_{ij} are normally and independently distributed with mean 0 and variance σ^2 :

$$\epsilon_{ij} \overset{i.i.d}{\sim} N(0, \sigma^2)$$

- *Fixed effects model*: a treatments represent the set of all possible treatments that we are interested in
- *Random effects model*: a treatments represent a random sample from a much larger set of treatments

Estimate model parameters

We will estimate the model parameters using least squares. To do this we calculate the error sum of squares, i.e.

$$\sum_{i=1}^a \sum_{j=1}^b (y_{ij} - \underbrace{\mu - \tau_i}_{-\mu_i})^2 = \sum_{i=1}^a \sum_{j=1}^b e_{ij}^2,$$

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon$$

and choose values for μ and τ_i that minimise this sum of squares.

The unique minimum is attained on

a + 1 eqns.
normal eqn.

$$\hat{\mu} = \bar{y}_{..}$$

$$\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{..} \quad \text{for } i = 1, \dots, a$$

where

$$\bar{y}_{..} = \frac{1}{a} \sum_{i=1}^a \frac{1}{n} \sum_{j=1}^n y_{ij}, \quad \bar{y}_{i\cdot} = \frac{1}{n} \sum_{j=1}^n y_{ij}.$$

Here $\bar{y}_{..}$ is the **grand mean** of all observations and $\bar{y}_{i\cdot}$ is the mean of i th treatment.

Analysis of Variance (ANOVA)

Treatment	1	2	...	a
	y_{11}	y_{21}	...	y_{a1}
y_{12}	y_{21}	y_{22}		y_{a2}
	$\vdots$	$\vdots$		$\vdots$
	y_{1n}	y_{1n}	...	y_{an}
Mean	$\bar{y}_{1.}$	$\bar{y}_{2.}$	...	$\bar{y}_{a.}$
St. Dev.	S_1	S_2	...	S_a

$$y_{ij} \quad i=1, \dots, a \\ j=1, \dots, n$$

Recall our definitions $\bar{y}_{..} = \sum_i \sum_j y_{ij} / an$, $\bar{y}_{i.} = \sum_j y_{ij} / n$.

ANOVA hypotheses

We are interested in detecting any differences between treatment means μ_i , hence we test the following hypothesis:

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_a$$

against the alternative

$$H_1 : \mu_i \neq \mu_k \text{ for at least one } i \neq k$$

Careful! The alternative hypothesis should be that at least two of the means differ, not that they are all different!

The primary tool when testing for equality of the means is based on a comparison of the variances within the groups and between the groups.

Variability decomposition

The ANOVA partitions the total variability in the sample data, described by the **total sum of squares**

$$SS_{\text{Tot}} = \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2 \quad (\text{df} = an - 1)$$

into the **treatment sum of squares** (= variability between groups)

$$SS_{\text{Tr}} = \sum_{i=1}^a n(\bar{y}_{i.} - \bar{y}_{..})^2 \quad (\text{df} = a - 1)$$

and the **error sum of squares** (= variability within groups)

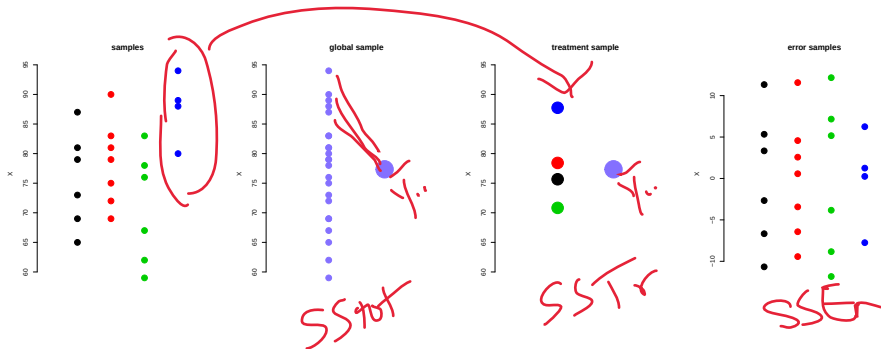
$$SS_{\text{Er}} = \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2 \quad (\text{df} = a(n - 1))$$

The sum of squares identity is

$$SS_{\text{Tot}} = SS_{\text{Tr}} + SS_{\text{Er}}$$

Variability decomposition

- SS_{Tot} : total amount of variation in the global sample
- SS_{Tr} : variation between the means of the treatment groups
- SS_{Er} : variation within groups



Mean Squared Error

The sample variance for the i th sample is $S_i^2 = \frac{1}{n-1} \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2$ which is an unbiased estimator for σ^2 : $\mathbb{E}(S_i^2) = \sigma^2$.

We know

$$SS_{\text{Er}} = \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2 = \sum_{i=1}^a (n-1)S_i^2,$$

hence

$$\mathbb{E}(SS_{\text{Er}}) = \sum_{i=1}^a (n-1)\mathbb{E}(S_i^2) = \sigma^2 \sum_{i=1}^a (n-1) = (an - a)\sigma^2$$

So another unbiased estimator for σ^2 is the **Mean Squared Error** MS_{Er}

$$\hat{\sigma}^2 = MS_{\text{Er}} = \frac{SS_{\text{Er}}}{an - a}$$

The number of degrees of freedom for this 'error' estimator of σ^2 is $a(n-1)$.

It can also be shown that $SS_{\text{Er}}/\sigma^2 \sim \chi^2_{an-a}$, so

$$\mathbb{P}\left(\frac{SS_{\text{Er}}}{\sigma^2} \geq \chi^2_{an-a,\alpha}\right) = 1 - \alpha$$

where $\chi^2_{an-a,\alpha}$ is the α -quantile of χ^2_{an-a} . Rearranging gives the $(1 - \alpha)\%$ confidence interval

$$0 < \sigma^2 \leq \frac{SS_{\text{Er}}}{\chi^2_{an-a,\alpha}}.$$

Treatment mean square

Now **if H_0 is true**, i.e if $\mu_1 = \mu_2 = \dots = \mu_a = \mu$, we have $\bar{y}_{i.} \sim \mathbb{N}(\mu, \frac{\sigma}{\sqrt{n}})$, that is $\sqrt{n}(\bar{y}_{i.} - \mu) \sim \mathbb{N}(0, \sigma^2)$, for all $i = 1, \dots, a$
 $\rightarrow \sqrt{n}(\bar{y}_{1.} - \mu), \sqrt{n}(\bar{y}_{2.} - \mu), \dots, \sqrt{n}(\bar{y}_{a.} - \mu)$, is a random sample whose sample variance

$$\frac{1}{a-1} \sum_{i=1}^a n(\bar{y}_{i.} - \bar{y}_{..})^2 = \frac{SS_{\text{Tr}}}{a-1}$$

is an unbiased estimator for $\sigma^2 \rightarrow$ the **Treatment Mean Square** MS_{Tr} , defined by

$$MS_{\text{Tr}} = \frac{SS_{\text{Tr}}}{a-1}$$

is also an **unbiased estimator for σ^2** .

The number of degrees of freedom for this 'treatment' estimator of σ^2 is $a - 1$.

ANOVA test

Thus we have two potential estimators of σ^2 :

- 1 MS_{Er} , which **always** estimates σ^2
- 2 MS_{Tr} , which estimates σ^2 **only when H_0 is true**

Then:

- if $MS_{Tr} \simeq MS_{Er}$, then H_0 is probably reasonable
- if $MS_{Tr} \gg MS_{Er}$, then H_0 should be rejected

(since if H_0 is not true, MS_{Tr} tends to exceed σ^2)

→ This will thus be a **one-sided** hypothesis test

We need to determine what “ $MS_{Tr} \gg MS_{Er}$ ” means so as to obtain a hypothesis test at given significance level α .

Sampling distribution

If H_0 is true, then SS_{Tr} follows a χ^2 distribution with $a - 1$ degrees of freedom and SS_{Er} follows a χ^2 distribution with $a(n - 1)$ degrees of freedom. The ratio

$$F = \frac{MS_{Tr}}{MS_{Er}} = \frac{\frac{SS_{Tr}}{a-1}}{\frac{SS_{Er}}{a(n-1)}}$$

follows a particular distribution known as the **Fisher's F -distribution** with $a - 1$ and $a(n - 1)$ degrees of freedom, which is usually denoted by

$$F \sim \mathbf{F}_{a-1, a(n-1)}$$

The Fisher's F -distribution

A random variable, say X , is said to follow Fisher's F -distribution with d_1 and d_2 degrees of freedom, i.e.

$$X \sim \mathbf{F}_{d_1, d_2}$$

if its probability density function is given by

$$f(x) = \frac{\Gamma((d_1 + d_2)/2)(d_1/d_2)^{d_1/2} x^{d_1/2-1}}{\Gamma(d_1/2)\Gamma(d_2/2)((d_1/d_2)x + 1)^{(d_1+d_2)/2}} \quad \text{for } x > 0$$

for some integers d_1 and $d_2 \rightarrow S_X = [0, +\infty)$ **Note:** the Gamma function is given by

$$\Gamma(y) = \int_0^{+\infty} x^{y-1} e^{-x} dx, \quad \text{for } y > 0$$

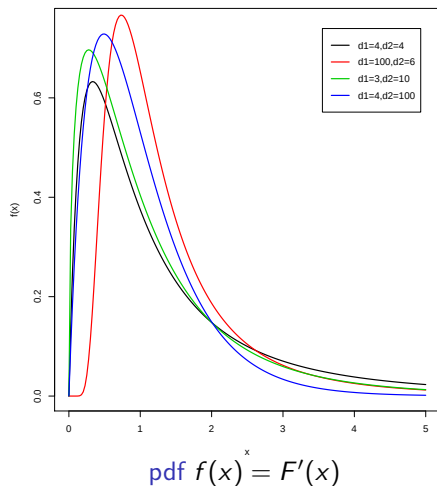
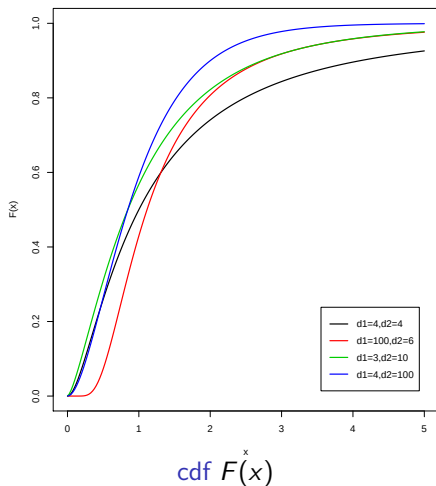
It can be shown that $\Gamma(y) = (y-1) \times \Gamma(y-1)$, so that, if y is a positive integer n ,

$$\Gamma(n) = (n-1)!$$

There is usually no simple expression for the F -cdf.

The Fisher's F -distribution

Some F -distributions



The Fisher's F -distribution: quantiles

We can define the **quantiles** of any F -distribution:

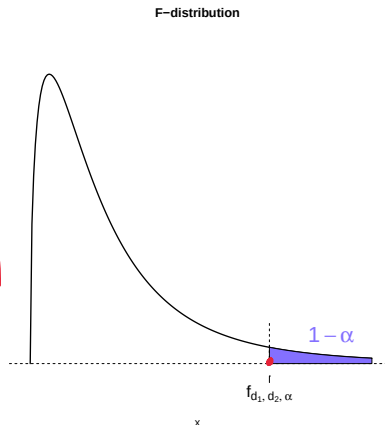
Let $f_{d_1, d_2; \alpha}$ be the value such that

$$\mathbb{P}(X > f_{d_1, d_2; \alpha}) = 1 - \alpha$$

for $X \sim \mathbf{F}_{d_1, d_2}$. The F -distribution is not symmetric, however it can be shown that

$$f_{d_1, d_2; \alpha} = \frac{1}{f_{d_2, d_1; 1-\alpha}}$$

off, ...



For any d_1 and d_2 , the main quantiles of interest may be found in the **F-distribution critical values tables**.

ANOVA test

The null hypothesis to test is $H_0: \mu_1 = \mu_2 = \dots = \mu_a$ versus H_a : not all the means are equal.

Evidence against H_0 is shown if $MS_{Tr} \gg MS_{Er}$, so we will reject H_0 whenever MS_{Tr} is much larger than MS_{Er} , i.e. $\frac{MS_{Tr}}{MS_{Er}}$ much larger than 1.

So for testing H_0 at significance level α , we need a constant c such that

$$\alpha = \mathbb{P} \left(\frac{MS_{Tr}}{MS_{Er}} > c \text{ if } H_0 \text{ is true} \right) \quad \alpha = 0.05$$

We know that, if H_0 is true, $F = \frac{MS_{Tr}}{MS_{Er}} \sim \mathbf{F}_{a-1, a(n-1)}$

$\Rightarrow$ we have directly that $c = f_{a-1, a(n-1); 1-\alpha}$ From observed values MS_{Tr} and MS_{Er} , the decision rule is:

① reject H_0 if $\frac{MS_{Tr}}{MS_{Er}} > f_{a-1, a(n-1); 1-\alpha}$

ANOVA test: p -value

The observed value of the test statistic is

$$f_0 = \frac{MS_{Tr}}{MS_{Er}}$$

and thus the p -value is given by

②

$$p = \mathbb{P}(F > f_0),$$

$1 - p^*(\dots)$

where $F \sim \mathbf{F}_{a-1, a(n-1)}$ ("the probability that the test statistic will take on a value that is at least as extreme as the observed value when H_0 is true")

From the F -distribution table, only bounds can be found for this p -value (use software to get an exact value)

This test is also often called the **F -test** or **ANOVA F -test**.

ANOVA table

SS
df

Source of Variation	Degrees of freedom	Sum of squares	Mean Square	F-statistic
Treatment	$a - 1$	$\sum_{i=1}^a n(\bar{y}_{i.} - \bar{y}_{..})^2$	$MS_{Tr} = \frac{SS_{Tr}}{a-1}$	$f_0 = \frac{MS_{Tr}}{MS_{Er}}$
Error	$a(n - 1)$	$\sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2$	$MS_{Er} = \frac{SS_{Er}}{a(n-1)}$	
Total	$an - 1$	$\sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2$		

Note: $df_{Tot} = df_{Tr} + df_{Er}$ and $SS_{Tot} = SS_{Tr} + SS_{Er}$

Example 1 in R

Let's perform ANOVA on example 1 using the following R code:

```
# Clear stored objects.
rm(list=ls())
# Setup data.
obs <- c(27,31,29,28,34,36,34,34,28,28,28,26,26,32,37,24,34,41,
        42,44,26,35,29,25,39,37,35,40,30,44,40,47,31,30,35,34,
        38,39,40,31,44,32,34,31,34,34,37,28,30,28,30,25,32,31,
        45,28,26,20,21,28) #response data
treatments <- rep(rep(c("task1","task2","task3","task4","task5",
"task6"), each = 2),5) #treatment data
pulse <- data.frame("pchange"=obs, "task"=treatments)
```

Example 1 in R

```
# Fit ANOVA model
fit <- aov(pchange~as.factor(task), data=pulse)
summary(fit) #display ANOVA table
```

#	Df	Sum Sq	Mean Sq	F value	Pr(>F)
# task	5	624.7	124.95	4.254	0.00247 **
# Residuals	54	1586.0	29.37		
# ---					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Handwritten notes: $P(F_{5,24} \geq 4.254) < 0.05$

The null hypothesis

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_6$$

or equivalently

$$H_0 : \tau_1 = \tau_2 = \dots = \tau_6 = 0$$

can be rejected with p-value 0.00247. We conclude that at least one task has a different effect on pulse rate than the others. In Lecture 4 we will introduce methods to determine which mean/means is/are different.

Example 1 in R

To obtain the estimated regression coefficients we use:

```
# Display regression coefficients (task1 as reference category).  
> coef(fit)
```

```
      (Intercept) as.factor(task)task2 as.factor(task)task3  
      32.7        -0.8          3.1  
as.factor(task)task4 as.factor(task)task5 as.factor(task)task6  
      5.2        -3.5        -3.6
```

```
mu_hat <- mean(pulse$pchange)
```

```
mu_hat
```

```
# 32.76667
```

```
mu_hat_i <- tapply(pulse$pchange, pulse$task, mean)
```

```
mu_hat_i
```

```
#task1 task2 task3 task4 task5 task6
```

```
#32.7 31.9 35.8 37.9 29.2 29.1
```

Example 1 in R

$$\hat{\tau}_i = \hat{\mu}_i - \hat{\mu}$$

```
tau_hat_i <- mu_hat_i-mu_hat
tau_hat_i
#      task1      task2      task3      task4      task5
#-0.06666667 -0.86666667  3.03333333  5.13333333 -3.56666667
# task6
#-3.66666667
```

Example 2

Suppose that you are planning to run an experiment with one treatment factor having three levels and no blocking factors. It has been determined that sample sizes $n_1 = 3$, $n_2 = n_3 = 5$. Assign at random 13 experimental units to the $a = 3$ treatments, so that the first treatment is assigned 3 units and the other two treatments are each assigned 5 units.

Example 2 in R

```
# Clear stored objects.
rm(list=ls())
# Overall sample size.
N <- 13
# Setup treatment vector.
trtmnt <- c(rep(1,3),rep(2,5),rep(3,5))
# Generate uniformly-distributed RVs.
set.seed(123) #to replicate results
rv <- runif(N)
# Collect data into data frame.
design <- data.frame(trtmnt,rv)
# Sort by rv.
design <- design[order(design$rv),]
# Assign experimental units.
design$expUnit <- 1:N
```

Example 2 in R

```
design
#   trtm      rv expUnit
#6   2 0.0455565      1
#1   1 0.2875775      2
#3   1 0.4089769      3
#12  3 0.4533342      4
#10  3 0.4566147      5
#7   2 0.5281055      6
#9   3 0.5514350      7
#13  3 0.6775706      8
#2   1 0.7883051      9
#4   2 0.8830174     10
#8   2 0.8924190     11
#5   2 0.9404673     12
#11  3 0.9568333     13
# Check if correct number assigned.
table(design$trtm)
# 1 2 3
# 3 5 5
```

Example 2 in R

```
# Extract experimental units receiving treatment 1.
design$expUnit[design$trtmt==1]
# [1]  2 3 9

# Extract experimental units receiving treatment 2.
design$expUnit[design$trtmt==2]
# [1]  1  6 10 11 12

# Extract experimental units receiving treatment 3.
design$expUnit[design$trtmt==3]
# [1]  4  5  7  8 13
```

Example 3

sulfamerazine		hemoglobin								
1	6.7	7.8	5.5	8.4	7.0	7.8	8.6	7.4	5.8	7.0
2	9.9	8.4	10.4	9.3	10.7	11.9	7.1	6.4	8.6	10.6
3	10.4	8.1	10.6	8.7	10.7	9.1	8.8	8.1	7.8	8.0
4	9.9	9.3	7.2	7.8	9.3	10.2	8.7	8.6	9.3	7.2

The data in the table above show the measurements of hemoglobin (grams per 100 ml) in the blood of brown trout. The trout were placed at random in four different troughs. The fish food added to the troughs contained, respectively, 0, 5, 10, and 15 g of sulfamerazine per 100 pounds of fish (coded 1, 2, 3, 4). The measurements were made on ten randomly selected fish from each trough after 35 days.

Example 3

(a) Assign at random 40 experimental units to the $a = 4$ treatments.

```
# Clear stored objects.
```

```
rm(list=ls())
```

```
# Overall sample size.
```

```
N <- 40
```

```
# Experimental unit vector.
```

```
expUnit <- 1:40
```

```
# Generate uniformly-distributed RVs.
```

```
set.seed(123) #to replicate results
```

```
rv <- runif(N) #generate uniformly distributed RVs
```

```
design <- data.frame(expUnit, rv)
```

```
# Sort by rv.
```

```
design <- design[order(design$rv),]
```

```
# Setup treatment vector and assign to experimental units.
```

```
sulfamerazine <- c(rep(1,10),rep(2,10),rep(3,10), rep(4,10))
```

```
design$sulfamerazine <- as.factor(sulfamerazine)
```

```
table(design$sulfamerazine)
```

```
#  1  2  3  4
```

```
# 10 10 10 10
```

Example 3

```
# Extract experimental units receiving sulfamerazine level 1.
design$expUnit[design$sulfamerazine==1]
# [1] 35 18 6 15 30 38 40 17 1 29
# Extract experimental units receiving sulfamerazine level 2.
design$expUnit[design$sulfamerazine==2]
# [1] 39 19 3 12 10 36 7 27 9 14
# Extract experimental units receiving sulfamerazine level 3.
design$expUnit[design$sulfamerazine==3]
# [1] 28 23 25 13 33 22 26 37 2 34
# Extract experimental units receiving sulfamerazine level 4.
design$expUnit[design$sulfamerazine==4]
# [1] 4 21 8 16 32 5 20 11 31 24
```

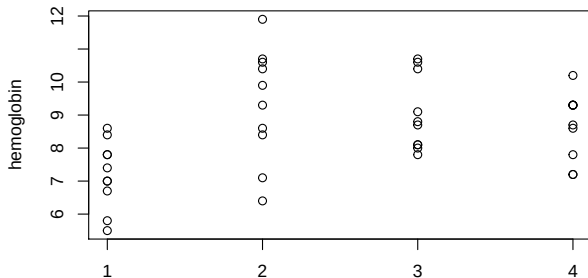
Example 3

(b) Plot the data and comment on the results.

```
# Input response data into data frame.
```

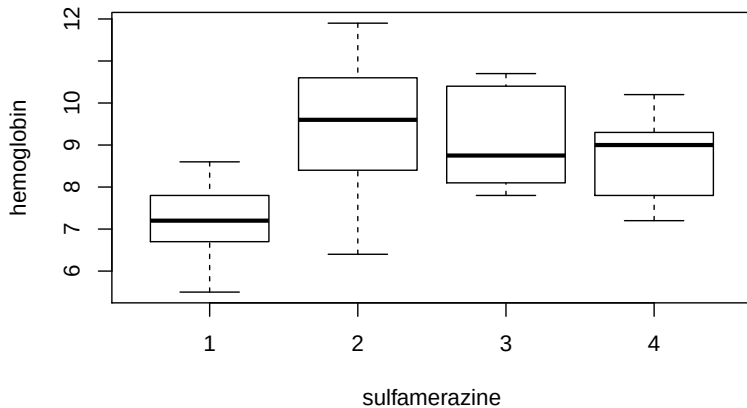
```
design$hemoglobin <- c(6.7,7.8,5.5,8.4,7,7.8,8.6,7.4,5.8,7,9.9,8.4,  
  10.4,9.3,10.7,11.9,7.1,6.4,8.6,10.6,10.4,8.1,10.6,8.7,10.7,  
  9.1,8.8,8.1,7.8,8,9.3,9.3,7.2,7.8,9.3,10.2,8.7,8.6,9.3,7.2)
```

```
stripchart(hemoglobin~sulfamerazine, vertical=TRUE, pch=1,  
  data=design)
```



Example 3

```
boxplot(hemoglobin~sulfamerazine, data=design) #boxplot
```



Example 3

(c) Write down a suitable model for this experiment.

$$\text{hemoglobin}_{ij} = \underbrace{\mu}_{\mu_i} + \tau_i + e_{ij}$$

$$i = 1, \dots, n$$
$$j = 1, \dots, 4$$

where

- hemoglobin_{ij} is hemoglobin level of the i -th trout receiving the j -th level of sulfamerazine
- $\mu_i := \mu + \tau_i = \mathbb{E}[\text{hemoglobin}_{ij}]$ is the population mean hemoglobin of the i -th level of sulfamerazine
- μ is the overall population mean hemoglobin level across all levels of sulfamerazine
- τ_i is the treatment effect on hemoglobin level of the i -th level of sulfamerazine
- e_{ij} is the random effect on hemoglobin level of the i -th trout receiving the j -th level of sulfamerazine.

Example 3

(d) Calculate the least squares estimate of the mean response for each treatment.

Calculate estimate of mu.

```
mu_hat <- mean(design$hemoglobin)
```

```
mu_hat
```

```
# [1] 8.5625
```

```
#
```

Calculate estimates of mu_i.

```
mu_hat_i <- tapply(design$hemoglobin, design$sulfamerazine, mean)
```

```
mu_hat_i
```

```
#      1      2      3      4
```

```
# 7.20 9.33 9.03 8.69
```

```
#
```

Calculate estimates of tau_i.

```
tau_hat_i <- mu_hat_i - mu_hat
```

```
tau_hat_i
```

```
#      1      2      3      4
```

```
# -1.3625  0.7675  0.4675  0.1275
```

$$\hat{\mu} = \bar{y}_{..}$$

$$\hat{\mu}_i$$

$$\hat{\tau}_i = \hat{\mu}_i - \hat{\mu}$$

Example 3

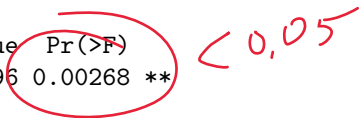
(e) Test the hypothesis that sulfamerazine has no effect on the hemoglobin content of trout blood.

```
# Fit ANOVA model.
```

```
fit <- aov(hemoglobin~sulfamerazine, data=design) #run ANOVA
```

```
summary(fit) #display ANOVA table
```

```
#           Df Sum Sq Mean Sq F value Pr(>F)
# sulfamerazine  3  26.80   8.934   5.696 0.00268 **
# Residuals     36  56.47   1.569
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



The null hypothesis

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_4$$

or equivalently

$$H_0 : \tau_1 = \tau_2 = \dots = \tau_4 = 0$$

can be rejected with p-value 0.00268. We conclude that at least one level of sulfamerazine has a different effect on hemoglobin than the others. In Lecture 4 we will introduce methods to determine which mean/means is/are different.

Example 3

(f) Calculate a 95% upper confidence limit for σ^2 .

See what components are available from aov object.

```
names(anova(fit))
```

```
# [1] "Df"      "Sum Sq"  "Mean Sq" "F value" "Pr(>F)"
```

```
#
```

Extract SSE.

```
sse <- anova(fit)$"Sum Sq"[2]
```

```
sse
```

```
# [1] 56.471
```

```
#
```

Calculate 95% upper bound.

```
rv <- qchisq(0.05, 36) #0.05-quantile from chi-squared distribution
```

```
sse/rv
```

```
# [1] 2.426918
```

$$= \frac{56.471}{36}$$

$$\frac{SSE}{n-a-1}$$

The estimate $\hat{\sigma}^2 = MSE = 1.569$ and the 95% upper bound on σ^2 is 2.426918.